

CONSERVATION

China's solar expansion policy reduces bird diversity

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Could the global transition to renewable energy create a green dilemma that pits carbon reduction against biodiversity conservation? This study examined the effect of policies promoting solar photovoltaics on local avian biodiversity using a panel dataset covering 2344 counties in China from 2014 to 2023. Policies that favored photovoltaic expansion led to reductions in bird diversity, disproportionately affecting wealthier and nondesert regions, as well as widespread species. The mechanism operated primarily through land conversion: Cropland and grassland were transformed into developed areas, reducing the diversity of vegetation. Paradoxically, the leaf area index increased, a pattern we term “inferior greening,” whereby diverse natural landscapes were replaced by dense but ecologically homogeneous vegetation. We argue that future photovoltaic development should be accompanied by strict biodiversity safeguards, especially in economically developed regions with high habitat complexity.

The global imperative to mitigate climate change has catalyzed an unprecedented expansion of renewable energy, with solar photovoltaics at the forefront. This transition, essential for decarbonization, can entail a profound transformation of land use, creating a potential conflict with biodiversity conservation. Nowhere is this dynamic more pronounced than in China, where the carbon neutrality strategy is driving a massive expansion of solar capacity projected to constitute nearly 45% of its energy mix by 2060 (1). Despite the global climate and local economic benefits, China's solar footprint reached an estimated 4520 km² in 2024 (2), equivalent in size to the US state of Rhode Island, placing pressure on terrestrial ecosystems. This policy-driven solar expansion raises an urgent and contentious question: Do these solar landscapes hasten the decline of avian populations, or can they be designed to function as ecological sanctuaries?

The question represents a classic “green dilemma” (3–5), and the scientific evidence remains divided. Some research highlights the negative impacts of photovoltaic installations, which alter and fragment landscapes (6), create visual and thermal disturbances (7, 8), and can lead to direct habitat loss for birds, threatening important biodiversity areas (9). Conversely, a growing body of work suggests potential cobenefits. The shaded, cooler, and more humid microclimates under photovoltaic panels can foster vegetation growth (10, 11), potentially enriching food sources, and the physical structures can act as “habitat islands” in degraded areas (12, 13). This ambiguity suggests that the ecological outcome is context-dependent, demanding strategic spatial planning to navigate trade-offs and identify low-conflict zones (3, 4, 14, 15).

In China, the location and scale of photovoltaic development are not merely functions of solar irradiation or land availability; they are

direct outcomes of a complex, multilayered institutional framework. This framework has evolved through successive “Five-Year Plans,” shifting focus over time from stimulating domestic manufacturing to promoting large-scale domestic deployment. This policy-driven expansion has been spatially uneven, shaped by targeted national initiatives. For instance, the Photovoltaic Poverty Alleviation Project, launched in 2014 and scaled up in 2016, explicitly linked solar deployment to economic aid by targeting 471 counties and approximately 35,000 villages (16). Concurrently, the designation of low-carbon pilot cities created another layer of policy-induced geographic variation in photovoltaic investment (17). These targeted policies, coupled with the natural concentration of solar resources in the Three-North region—an expanse spanning northwest, northern, and northeast China that is characterized by desertification, evaporation, and abundant solar irradiance—have forged a distinctive, quasi-experimental landscape where the stringency of policy support for photovoltaic varies across counties. Therefore, analyzing the ecological consequences of photovoltaic expansion without accounting for these policy drivers constitutes an important research gap.

To fill this knowledge gap, we investigated how the stringency of county-level photovoltaic policies affected bird diversity across China. Our study makes three principal contributions to the fields of energy policy and conservation science. First, by shifting the analytical focus from installed capacity to policy stringency, we shed light on the institutional drivers of the energy-biodiversity nexus. Much of the existing literature has quantified the ecological consequences of renewable energy by correlating biodiversity metrics with the physical footprint or capacity of existing infrastructure (18, 19). This approach, while valuable, analyzes the symptoms of landscape change rather than its underlying institutional causes. Our study moved upstream to assess how the stringency of the policy environment itself—the primary driver of the scale and location of development—shapes ecological outcomes. This aligns with calls for a deeper understanding of policy conflicts and synergies in conservation (20) and provides a more direct, actionable framework for evaluating the true socioecological costs of different renewable energy strategies.

Second, our analysis embeds the study of solar impacts within China's distinctive and heterogeneous institutional landscape. The ecological effects of renewable energy are known to be highly context-dependent, varying with local geography, land-use history, and policy frameworks (3). Although recent studies have begun to explore renewable energy's impacts in China (21, 22), they have not systematically dissected how outcomes differ across critical sociopolitical strata. By comparing the varying ecological burdens across poverty versus non-poverty counties and across distinct geographical zones (such as arid and semiarid regions, particularly the desert and Gobi areas of northern China versus nondesert areas), our study aims to evaluate the limitations of one-size-fits-all conservation mandates. This comparative approach highlights the necessity of region-specific frameworks to ensure that standardized solar deployment models are ecologically sustainable across diverse geographic landscapes.

Third, our mediation analysis uncovered a counterintuitive mechanism we term “inferior greening,” which challenges conventional interpretations of ecological restoration indicators. Although satellite-derived greenness metrics such as leaf area index (LAI) are often used as proxies for ecosystem health, we reveal that policy-driven photovoltaic expansion was associated with an increase in LAI that simultaneously mediates a reduction in bird diversity. Rather than indicating

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ecological vitality, this pattern highlights a potential decoupling of greenness and biodiversity, masking underlying habitat loss. The dense but homogeneous undergrowth that may flourish under solar panels lacks the structural complexity and resource diversity of the natural habitats it replaces, thereby failing to support specialist avian communities (23). Our investigation of this mechanism calls for a rethinking of

what constitutes ecological health in the context of renewable energy expansion and underscores the danger of relying on simplistic biomass indicators for environmental monitoring.

Results

To investigate the impacts of solar energy expansion on ecosystems, we constructed a comprehensive panel dataset covering 2344 Chinese counties from 2014 to 2023 (46,528 county-month observations), integrating policy, biodiversity, and environmental variables (tables S1 to S5). Bird observation data were sourced from the China Birdwatching Record Center, a high-precision citizen science platform (22, 24), with species conservation statuses derived from the 2021 List of State Key Protected Wild Animals. To capture the institutional drivers of solar deployment, we systematically web-scraped official photovoltaic policy documents across provincial, municipal, and county government portals. Bird diversity was quantified using the Shannon index, which integrates species richness and evenness to reflect community complexity and health (25). Lastly, we developed a photovoltaic policy stringency index (PSI) by evaluating the scraped texts. Policies were

Table 1. Solar policy stringency significantly reduces bird diversity. This table presents the association between bird diversity and the strength of photovoltaic policies in 2344 counties from 2014 to 2023. The dependent variable is the Shannon index. The independent variable is PSI, the weighted sum of solar-related policy documents at provincial, municipal, and county levels. Controls include annual average temperature (Temp), annual average wind speed (Wind), the logarithm of population density (Pop), the logarithm of bird-watching duration per birdwatcher (Duration), the logarithm of carbon emissions (Carbon), the proportional area of water bodies (Water), forests and shrubs (Green), farmland (Farm), and grassland (Grass). All specifications include county and year-month fixed effects (FE). Robust standard errors clustered at the county level are reported in parentheses. ***, **, * denote significance at the 1, 5, and 10% levels, respectively. R^2 , coefficient of determination.

	(1) ShannonBD	(2) ShannonBD	(3) ShannonBD
PSI	−0.0157*** (0.0037)	−0.0127*** (0.0036)	−0.0125*** (0.0037)
Temp		−0.0099 (0.0180)	−0.0073 (0.0181)
Wind		0.0479 (0.0735)	0.0435 (0.0736)
Pop		−0.0998 (0.1287)	−0.0885 (0.1287)
Duration		0.1639*** (0.0057)	0.1638*** (0.0057)
Carbon		−0.1007** (0.0421)	−0.1118*** (0.0431)
Water			−1.2799 (1.1270)
Green			1.1518 (0.9519)
Farm			0.0657 (0.7088)
Grass			1.2642 (0.9469)
Year-month FE	YES	YES	YES
County FE	YES	YES	YES
Observations	46,371	46,371	46,371
R^2	0.2715	0.2998	0.2999

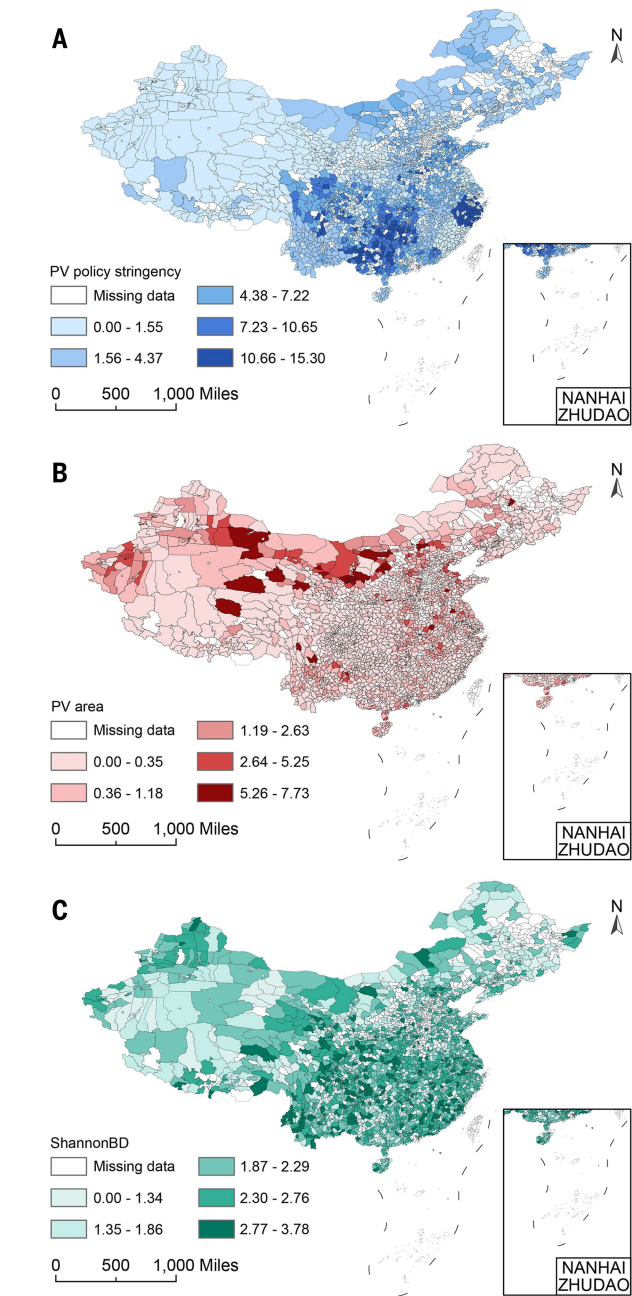


Fig. 1. The spatial distribution of solar energy expansion and bird diversity across China. The figure presents county-level choropleth maps illustrating the geographic patterns of key variables. White areas indicate missing data. (A) Photovoltaic policy stringency index (PSI), measured as the weighted sum of policy documents at provincial, municipal, and county levels. (B) The total area of centralized photovoltaic power stations within each county. (C) Bird diversity (ShannonBD), calculated as the negative sum of the proportion of each bird species multiplied by its natural logarithm. Nanhai Zhudao, South China Sea Islands.

scored from 1 to 9 according to administrative binding force, ranging from broad outlines to codified regulations, and weighted by the issuing authority's administrative tier to reflect the nested structure of policy implementation (table S3). Spatial patterns and temporal dynamics (2014–2023) of the photovoltaic PSI, power station deployment, and bird diversity exhibit distinct county-level distributions (Fig. 1 and fig. S1). **The PSI correlates positively with the actual land area of centralized photovoltaic plants across counties ($\beta = 0.0222$, standard error (SE) = 0.0026, $P < 0.01$; fig. S2 and table S6), and the correlation appears to be the strongest with a 2-year lag, capturing the time needed from policy incentive to project completion. To address endogeneity, we used historical sunshine interacting with the inverse of climate policy uncertainty as an instrumental variable (IV), which shows a strong partial correlation with the PSI ($\beta = 0.0401$, SE = 0.0028, $P < 0.01$) and confirms the instrument's relevance (fig. S3).**

We estimated the relationship between policy stringency and bird diversity using a high-dimensional, two-way fixed-effects regression model. This approach isolates the policy effect by controlling for time-invariant county characteristics and common temporal shocks, alongside time-varying meteorological, geographic, and socioeconomic covariates (e.g., temperature, wind speed, population density, bird-watching duration, carbon emissions, and land cover proportions). The model includes county fixed effects to absorb time-invariant unobserved heterogeneity, such as fundamental geographic characteristics, and year-month fixed effects to account for seasonality and aggregate temporal trends.

Intensive solar policies linked to local bird diversity decline

Our analysis indicates a negative relationship between solar policy stringency and local bird diversity (Table 1). **Specifically, a one-standard-deviation increase in policy stringency corresponds to a 2.10% reduction in the Shannon index ($\beta = -0.0125$, SE = 0.0037, $P < 0.01$).** This finding aligns with observations that utility-scale solar facilities necessitate land-use changes that alter local habitats, often leading to habitat degradation and fragmentation (26–28). Such land conversions can indirectly affect avifauna by displacing species and altering resource availability (29). **This core finding remains robust across a comprehensive suite of alternative specifications, including instrumental-variable and propensity-score matching approaches to address endogeneity,** sensitivity tests using alternative sampling criteria, and exclusion of county-month observations below minimum bird-watching record thresholds, as well as tests utilizing alternative biodiversity and policy metrics, controlling for concurrent environmental policies, and accounting for spatial and seasonal dynamics (tables S7 to S16).

Disproportionate effects across vulnerable regions and species

The aggregate relationship between solar policy stringency and bird diversity exhibits spatial and socioeconomic variation (Fig. 2 and tables S17 and S18), highlighting that the ecological outcomes of renewable energy are context-dependent. The reduction in bird diversity is more pronounced in nonpoverty counties ($\beta = -0.0132$, SE = 0.0047, $P < 0.01$) compared with poverty-designated counties. A one-unit increase in policy stringency within nonpoverty counties

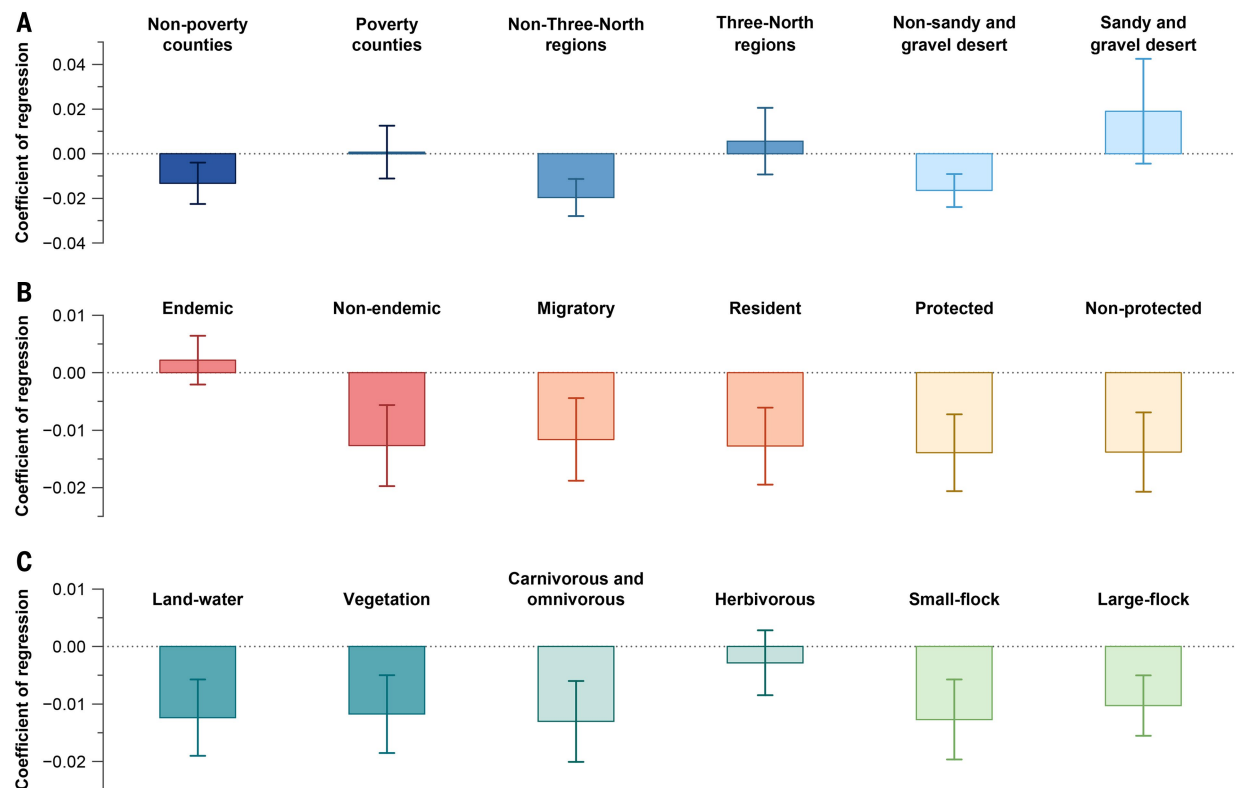


Fig. 2. Mapping the uneven toll of solar expansion on vulnerable groups. The figure shows regression coefficients from heterogeneity analyses for the period 2014–2023. Error bars represent 95% confidence intervals. **(A)** Cross-sectional heterogeneity by geographic and socioeconomic characteristics: poverty status (nonpoverty versus poverty counties), regional location (non-Three-North versus Three-North regions), and land surface type (nonsandy and gravel desert versus sandy and gravel desert). **(B)** Heterogeneity by species-level ecological characteristics: endemism (endemic versus non-endemic), migratory status (migratory versus resident), and conservation status (state-protected versus nonprotected). **(C)** Heterogeneity by species-level functional traits: nesting guilds (vegetation nesters versus land-water nesters), dietary guilds (carnivorous and omnivorous versus herbivorous), and flocking behavior (small-flock versus large-flock).

corresponded to a 0.54% decline in bird diversity relative to the sample mean, translating to a 2.22% decrease per standard-deviation increase.

The negative association is observed primarily outside the Three-North region ($\beta = -0.0196$, $SE = 0.0042$, $P < 0.01$) and in nondesert areas ($\beta = -0.0165$, $SE = 0.0038$, $P < 0.01$). By contrast, within the Three-North region and sandy and gravel desert regions—areas characterized by high solar irradiance but naturally sparse vegetation and lower baseline habitat complexity—the effect of photovoltaic policy stringency on bird diversity is not significant. The avifauna in these arid landscapes may already be adapted to open environments, making them less sensitive to the marginal habitat changes introduced by solar farms. This suggests that the conversion of land for solar infrastructure has a more measurable impact on avian communities in regions with historically higher ecosystem complexity and productivity.

Species vulnerability varied significantly. Photovoltaic policies negatively impacted non-endemic species ($\beta = -0.0127$, $SE = 0.0036$, $P < 0.01$), whereas China’s endemic species, often habitat specialists concentrated in mountainous forests unsuitable for solar development, were unaffected (table S18). Both migratory ($\beta = -0.0116$, $SE = 0.0037$, $P < 0.01$) and resident birds ($\beta = -0.0128$, $SE = 0.0034$, $P < 0.01$) suffered significant declines. A one-unit increase in policy stringency decreased the Shannon index by 0.47% for migratory species. Whereas resident species face continuous pressure from permanent habitat conversion, migrants lose essential staging areas, turning localized land-use changes into flyway-scale survival threats. Notably, regions with aggressive solar policies spatially coincide with these critical migratory corridors (fig. S4). Furthermore, photovoltaic policy stringency negatively correlated with the diversity of both state-protected ($\beta = -0.0139$, $SE = 0.0034$, $P < 0.01$; a 0.57% reduction per unit increase) and nonprotected species ($\beta = -0.0138$, $SE = 0.0035$, $P < 0.01$; a 2.32% decline per standard-deviation increase), underscoring the necessity of integrating broad biodiversity conservation into photovoltaic project planning.

The functional traits of bird species mediated their vulnerability to solar farm development. Nesting strategy was an important predictor: Species that nested in vegetation experienced a negative effect with a coefficient of -0.0118 ($SE = 0.0035$, $P < 0.01$), compared with those that nested on the ground or water surfaces ($\beta = -0.0124$, $SE = 0.0034$, $P < 0.01$). Dietary guilds likewise influenced sensitivity. We detected a significant negative association between PSI and carnivorous and omnivorous species ($\beta = -0.0130$, $SE = 0.0036$, $P < 0.01$), whereas herbivorous species showed no significant response. The impacts were further modulated by flocking behavior. Heterogeneity analyses indicated that intensive photovoltaic policies significantly reduced the diversity of both small-flocking ($\beta = -0.0127$, $SE = 0.0035$, $P < 0.01$) and large-flocking birds ($\beta = -0.0103$, $SE = 0.0027$, $P < 0.01$). The large-scale deployment of solar arrays driven by these policies often results in severe habitat fragmentation and the loss of essential foraging and breeding grounds, threatening avian populations regardless of flock size. Collectively, these results indicated that nesting substrate, trophic position, and social foraging strategies were key functional traits determining bird vulnerability to solar farm development.

Vegetation, human activity, and land-use conversion as causal pathways

To understand the pathways linking policy stringency to changes in bird diversity, we examined intermediate environmental variables (Table 2). Utility-scale solar facilities alter the physical structure of habitats, which is a key determinant of species diversity according to the habitat heterogeneity hypothesis (23, 30). Satellite-derived data show that photovoltaic policy stringency is negatively associated with the normalized difference vegetation index (NDVI) ($\beta = -0.0609$, $SE = 0.0138$, $P < 0.01$), reflecting the initial clearing of

Table 2. Statistical evidence for the inferior greening mechanism. This table presents results of the mechanism tests. The dependent variables in columns (1), (2), and (3) are the normalized difference vegetation index (NDVI), nighttime light intensity (Nightlight), and leaf area index (LAI), respectively. The independent variable is PSI. Controls and fixed-effects structure are as shown in Table 1. Robust standard errors clustered at the county level are reported in parentheses. ***, **, * denote significance at the 1, 5, and 10% levels, respectively.

	(1) NDVI	(2) Nightlight	(3) Leaf area
PSI	−0.0609*** (0.0138)	−0.2220*** (0.0256)	0.0039*** (0.0007)
Temp	−0.9656*** (0.0842)	0.0384 (0.1017)	0.0280*** (0.0031)
Wind	−3.1301*** (0.3847)	−0.2611 (0.4120)	−0.0432*** (0.0132)
Pop	−0.4924 (0.6068)	−0.9999 (0.9217)	−0.0132 (0.0159)
Duration	0.0221** (0.0103)	0.0604*** (0.0142)	0.0007 (0.0004)
Carbon	0.0132 (0.2187)	0.0315 (0.2745)	−0.0061 (0.0059)
Water	5.4642 (3.8063)	−88.3865*** (14.2655)	0.2704** (0.1155)
Green	17.3088*** (3.8053)	−18.2376* (10.2519)	0.3453** (0.1434)
Farm	8.2027*** (3.1254)	−32.0623*** (8.4823)	0.0844 (0.0918)
Grass	−7.5378 (5.5264)	−25.7797*** (9.1801)	0.0136 (0.1366)
Year-month FE	YES	YES	YES
County FE	YES	YES	YES
Observations	46,371	46,371	46,338
R ²	0.9917	0.9915	0.9921

native vegetation. Concurrently, policy stringency is positively associated with the LAI ($\beta = 0.0039$, $SE = 0.0007$, $P < 0.01$). This combination suggests a phenomenon we term inferior greening: a transition from structurally diverse native ecosystems to denser, but ecologically homogeneous, understory vegetation beneath solar panels, driven by altered microclimates (31). From a management perspective, this shift may reflect compliance strategies in which fast-growing monoculture species are utilized to satisfy basic environmental restoration mandates. This structural degradation explains the specific vulnerability of vegetation-nesting birds, which rely on complex plant architecture for concealment, and carnivorous or omnivorous birds, which depend on the invertebrate prey base supported by native flora. Herbivorous birds, conversely, may adapt to the dense ground cover, explaining their lack of significant decline. Beyond direct habitat loss, the large-scale conversion of land into industrial energy landscapes exacerbates habitat fragmentation, isolating natural patches and disrupting avian movement (32). Furthermore, these facilities may act as ecological traps through the “lake effect,” in which polarized light reflection causes birds to mistake panels for water bodies, leading to collision mortality (33). We also tested the hypothesis that artificial light at night (ALAN) from industrial presence drives the diversity reduction (34). However, policy stringency is negatively correlated with nighttime light intensity ($\beta = -0.2220$, $SE = 0.0256$, $P < 0.01$), likely because large-scale solar projects are situated in remote, low-light areas. This indicates that direct habitat alteration and fragmentation,

rather than light pollution, are the primary mechanisms connecting policy to changes in bird diversity.

Socioecological consequences

Further analysis delineated the temporal and broader land-use implications of solar policy stringency (Fig. 3 and table S19). By separating the immediate implementation of new policies (“flow”) from the accumulated history of past policies (“stock”), we found that the contemporaneous policy flow significantly reduces bird diversity ($\beta = -0.0105$, $SE = 0.0044$, $P < 0.05$), whereas the lagged policy stock does not. This suggests that the initial phases of habitat conversion driven by new mandates are the primary source of ecological change. In addition to ecological changes, we observed land-use competition affecting agriculture; a one-unit increase in photovoltaic policy stringency is associated with a 2.88% decline in agricultural crop yield ($\beta = -0.0288$, $SE = 0.0081$, $P < 0.01$). This reduction is independent of the decline in avian biodiversity, pointing to direct spatial competition between solar infrastructure and agricultural production. **Lastly, photovoltaic policy stringency alters the structure of avian communities by reducing overall species richness ($\beta = -0.9751$, $SE = 0.1828$, $P < 0.01$) while slightly increasing species evenness ($\beta = 0.0018$, $SE = 0.0007$, $P < 0.01$).** This indicates an environmental filtering process in which habitat simplification removes specialized species, leaving a community with fewer total species but a more balanced relative abundance among the remaining generalists.

Discussion

Transitioning to renewable energy is imperative for global climate mitigation and remains the cornerstone of China’s dual carbon strategy. However, our findings highlight a critical green dilemma: Without meticulous spatial planning, policy-driven solar expansion can inadvertently compromise local biodiversity. Resolving this conflict is not about restricting solar development, but rather optimizing it. Our results provide a data-driven, scientific basis for refining photovoltaic

deployment, ensuring that the renewable energy transition achieves a win-win scenario for both high-quality economic development and ecological conservation.

First, spatial differentiation is paramount for future policy design. Large-scale utility photovoltaic should be strategically concentrated in the arid and semiarid regions, particularly the desert and Gobi areas of northern China. These areas offer a path of least ecological resistance, where baseline habitat complexity is low and solar infrastructure causes minimal ecological displacement. Conversely, in wealthier, non-desert regions with productive ecosystems, land conversion for utility-scale photovoltaic should be strictly minimized. To meet energy demands in these sensitive areas, policies should incentivize ecofriendly distributed solar, such as rooftop installations and agrivoltaic systems. These alternatives maintain vertical habitat structures and safeguard agricultural food security, directly mitigating land-use conflicts.

Second, the administrative implementation of renewable energy mandates should transition from abrupt policy shocks to phased, well-integrated rollouts. Our analysis indicates that immediate policy “flows” drive rapid habitat alteration. Therefore, future solar directives must embed mandatory biodiversity impact assessments and ecological safeguards directly into the initial stages of policy design. Establishing dynamic buffer zones and integrating conservation planning before project execution will prevent hasty, ecologically disruptive land clearing, all while maintaining the necessary momentum of the clean energy transition.

Lastly, environmental monitoring and restoration standards for solar facilities must be upgraded to address the phenomenon of inferior greening. Relying solely on standard biomass indicators, such as the LAI, can mask the severe structural degradation of native habitats. Regulatory frameworks should redefine ecological compliance by mandating the cultivation of diverse, native flora beneath and around solar arrays, explicitly prohibiting the reliance on homogeneous, low-cost ground cover. These ecological restoration standards should be tailored to reflect spatial heterogeneity, ensuring that mitigation

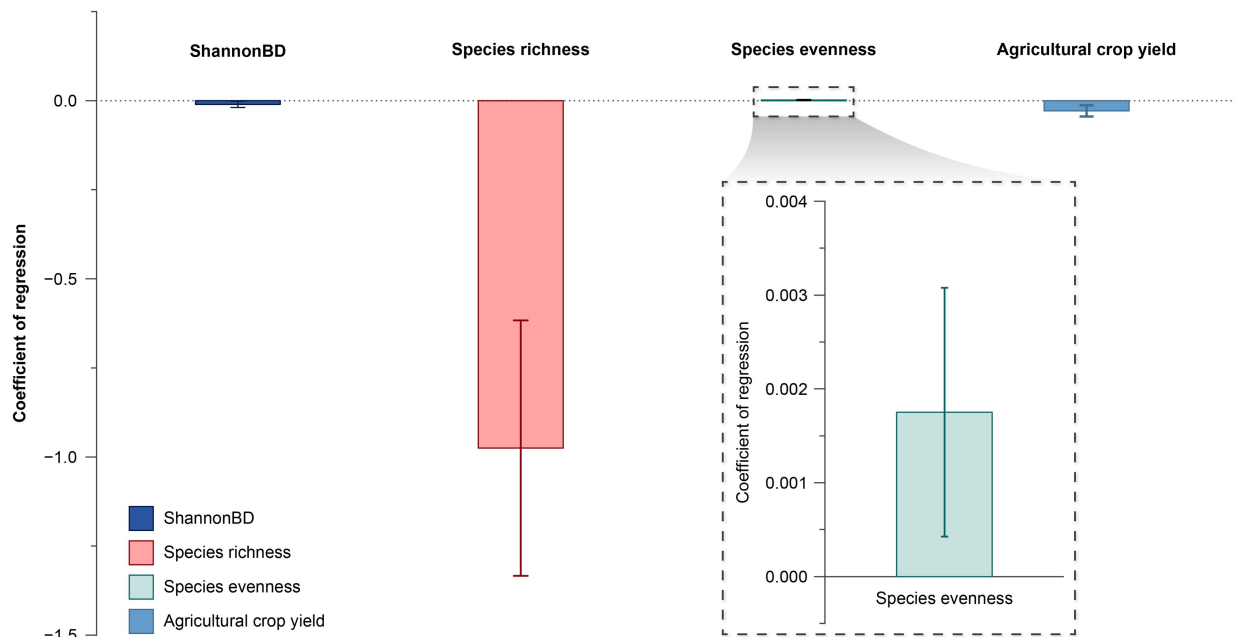


Fig. 3. Socioecological consequences of solar energy expansion. The figure displays regression coefficients of PSI on four dependent variables; error bars represent 95% confidence intervals. The first bar shows the effect of PSI on the Shannon index after controlling for one-period-lagged PSI. The second bar shows the effect of PSI on species richness (the total number of species observed within the survey area). The third bar shows the effect of PSI on species evenness (the ratio of the Shannon index to the logarithm of species richness), with an inset providing a zoomed-in view owing to the small magnitude of the coefficient. The fourth bar shows the effect of PSI on agricultural crop yield (one-period lead, log-transformed).

strategies are context-specific for desert versus nondesert biomes. By introducing financial incentives for biodiversity-positive solar farm management, policy-makers can encourage developers to restore functional, multitiered habitats that support complex avian food webs.

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SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.aee0747
Materials and Methods; Supplementary Text; Figs. S1 to S4; Tables S1 to S19; References (35–71); MDAR Reproducibility Checklist
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China's solar expansion policy reduces bird diversity

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Editor's summary

The world needs to develop clean, renewable energy. However, given the enormous human population and our energy needs, clean energy also will have a large footprint on nature. Solar energy represents one of the cleanest forms of renewables, but solar farms can greatly alter the environment. Zhang *et al.* looked at the effects of solar power-promoting policies on bird diversity across over 2000 counties in China (see the Perspective by Liang). They found that counties with the most pro-photovoltaic policies had reductions in bird diversity, primarily due to land conversion. They argue that solar developments must be proactive about countering such losses, especially focusing siting efforts in regions such as deserts, where vegetation alteration can be minimized. —Sacha Vignieri

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